



DataLake3Auth: Secure Offline Field Authentication

Hackathon 7.0 Definitive Solution

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The Hostile Environment: Defining the Physical Constraints



Zero-Network Zones

Remote sites lack usable coverage. Standard cloud face-matching APIs fail outright outright where Datalake 3.0 crews actually operate.



Rampant Attendance Fraud

Naive facial recognition is easily defeated. Buddy punching utilizing high-resolution printed photographs or screen replays demands robust, hardware-independent anti-spoofing.



Constrained Device Budgets

Authentication must occur on typical mid-range hardware: Android 8+ or iOS 12+ devices, limited to 3 GB RAM, with strictly CPU-only processing capabilities.

The Hackathon 7.0

Non-Negotiables

- > **Platform:** React Native, cross-platform Android + iOS.
- > **Performance:** > 95% recognition accuracy on established benchmarks.
- > **Security:** Fully offline liveness detection requiring active prompts (blink, smile, head turn).
- > **Architecture:** Open-source technologies only.
- > **Lifecycle:** Sync to AWS when network returns, then completely purge local data.
- > **Efficiency:** Complete model payload strictly ~20 MB. Verification speed < 1s per user.

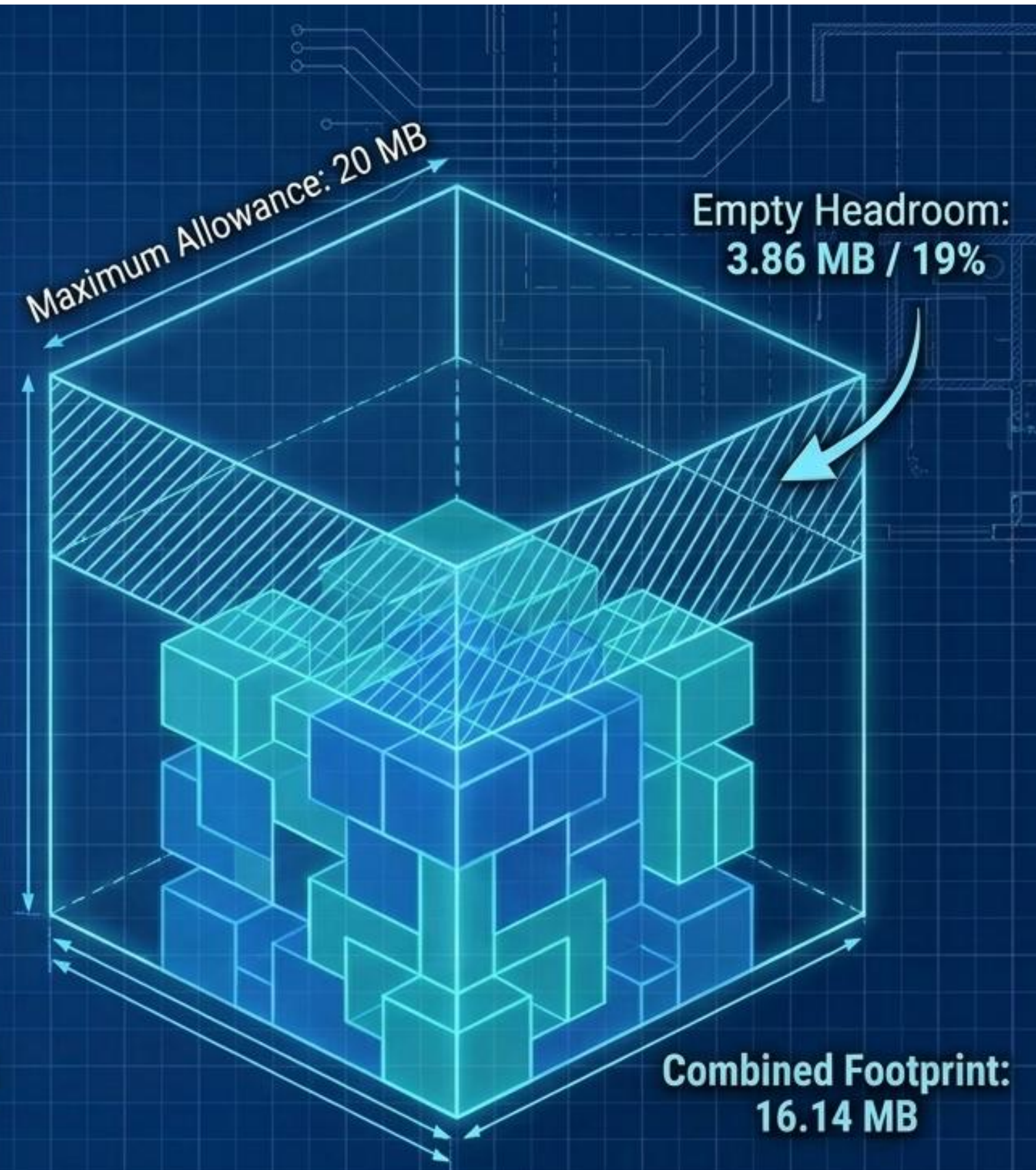
Constraint Compliance Matrix

Parameter	Brief Requirement	DataLake3Auth Execution	Headroom
Accuracy	Target > 95%	✔ 97.28% (LFW protocol)	+2.28% margin
Memory Footprint	Target < 20 MB	⚡ 16.14 MB	19% under budget
Latency	Target < 1,000 ms	📶 783 ms avg (16.84 ms embedding)	~200 ms faster
Hardware Base	Target CPU-only, 3GB RAM	Quantised lightweight CNNs (No GPU required)	Universal compatibility
Ecosystem	Target Open Source	Open models + runtime; public GitHub repo	Confirmed

Footprint Engineering: 19% Under Budget

Achieving a sub-20 MB footprint required architectural discipline, avoiding massive general-purpose models in favor of tuned, task-specific networks.

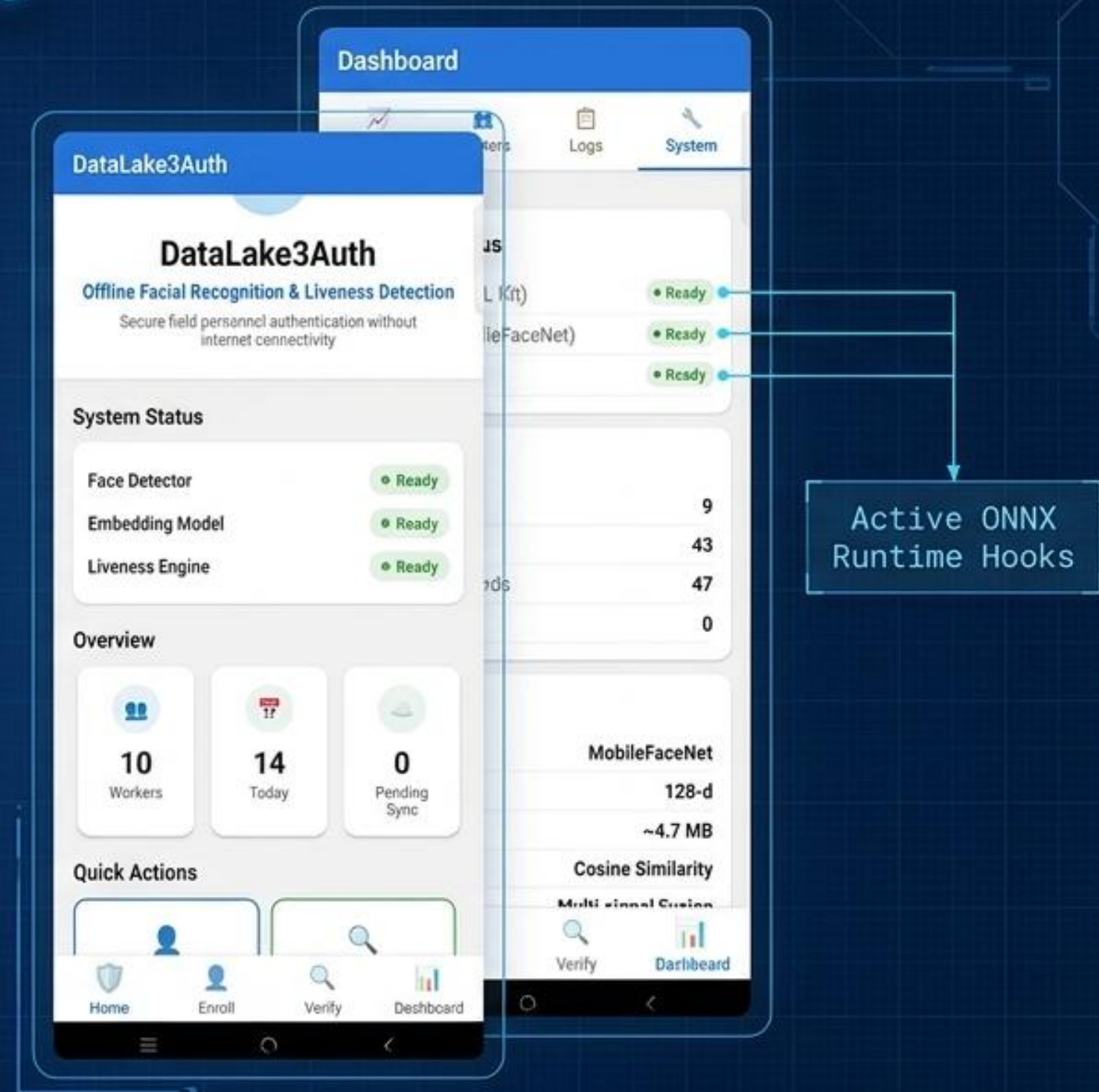
- > **Shared Detector:** One lightweight ONNX detector handles face detection, landmarks, eye/smile cues, AND head pose. One model feeds both alignment and liveness.
- > **Compact Embedder:** MobileFaceNet-class CNN utilizes global depthwise and separable convolutions to remain at just ~4.7 MB.
- > **Quantisation:** ONNX weight quantisation strictly reduces payload while maintaining 97.28% accuracy.



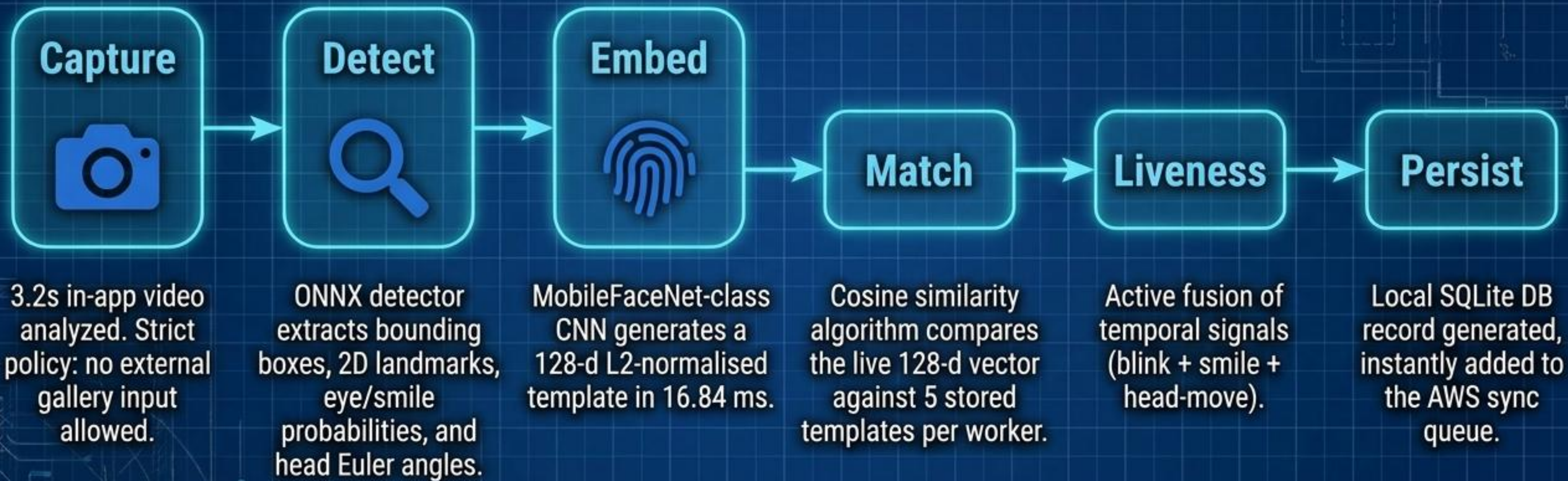
The Edge-Native User Experience

A complete React Native application orchestrating **Enrolment**, **Verification**, and **System Management** entirely on-device via a native ONNX inference bridge.

- **Zero Internet Calls:** Verification occurs completely independently of network status.
- **System Status Telemetry:** Runtime configurations, including Face Detector, Embedding Model, and Liveness Engine, are actively monitored and inspectable in-app.
- **Cross-Platform Architecture:** RN prototype interfaces with an optimized C++/native module bridge for maximum CPU utilization.



The Offline Processing Pipeline



The Anti-Spoofing Philosophy: Time as a Filter

Why do high-resolution photos and screen replays fail against DataLake3Auth?

Static Attack (Photo / Replay)



A perfect 2D photograph of an enrolled worker will generate a high cosine similarity match, but it produces absolutely zero temporal delta. Without movement across frames, liveness fails by design.

Dynamic Subject (3D Reality)



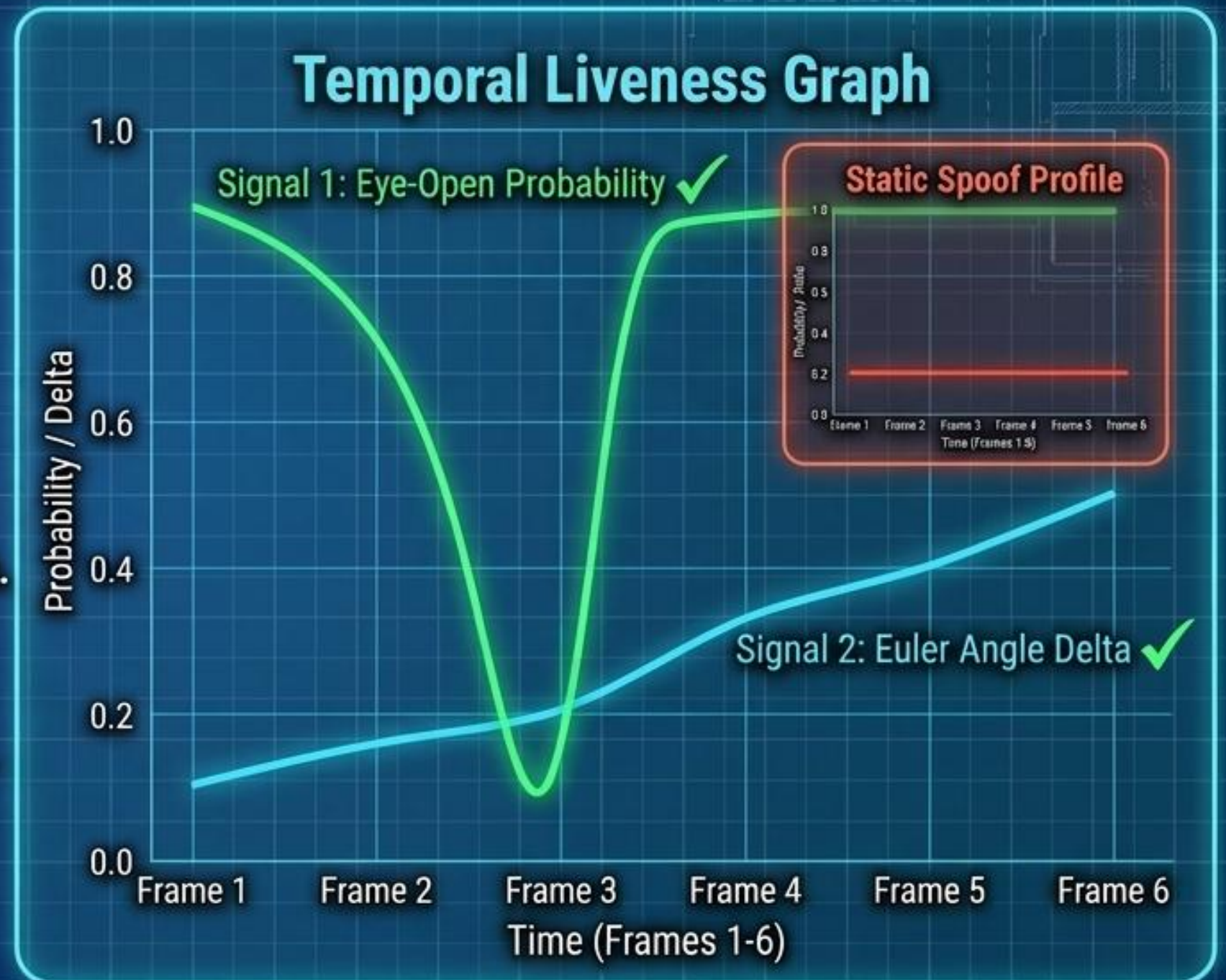
The system analyzes 6 distinct video frames over a 3.2-second window.

In-App Capture Only: System refuses pre-recorded video or gallery uploads. The clip is guaranteed volatile.
Guided Prompts: Users are instructed to 'blink or turn your head slightly' during capture.

Multi-Signal Temporal Fusion Dashboard

Detecting 3D humanity via mathematical deltas across a 3.2-second window.

- **Signal 1: The Blink.** The ONNX detector tracks eye-open probability. A valid blink manifests as a distinct, rapid dip and immediate recovery curve.
- **Signal 2: Head Movement.** Euler angles (pitch, yaw, roll) are tracked. A live subject produces continuous, micro-fluctuations in these angles, confirming 3D spatial presence.
- **Signal 3: The Smile.** Changes in smile probability serve as an active secondary cue.



The Two-Gate Verification Synthesis

An attendance record is only saved to the local database if both independent **cryptographic** and **temporal** conditions are met simultaneously.

Gate 1: Mathematical Identity

The live 128-d embedding's cosine match must clear the 95% threshold, with a sufficient margin (e.g., 0.467) over the next closest runner-up. 

Gate 2: Temporal Liveness

The fusion score derived from blink and head-movement deltas across 6 frames must pass the designated liveness threshold (e.g., 72%).

AND

Attendance Logged



Verified: Divyaman

Result: 783 ms average full verification latency.

Robust State: The Enrollment Workflow

A streamlined four-step wizard generates the baseline required for 97.28% offline accuracy.

1 Details

- Capture worker Name, ID, Department, and Role.

2 Capture

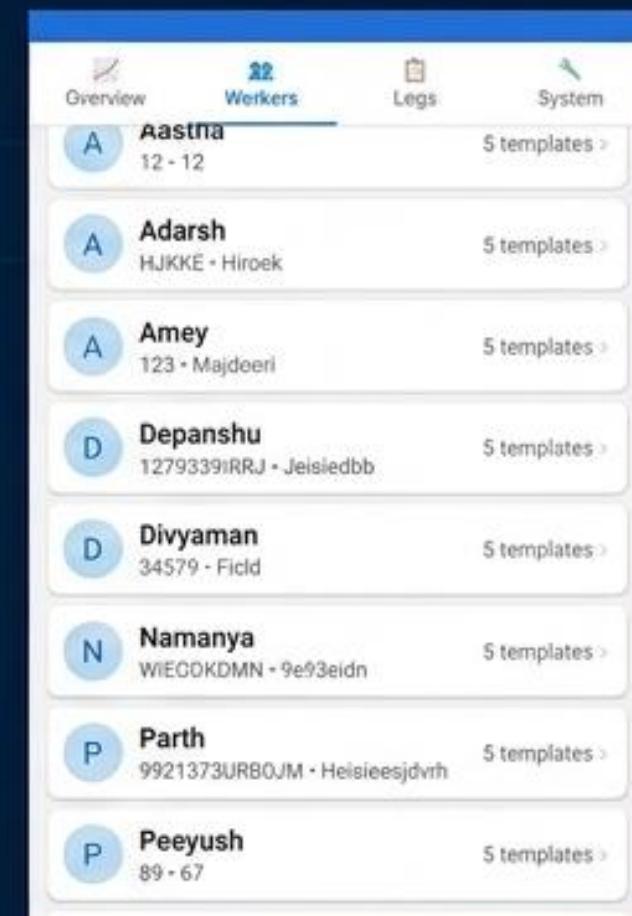
- Record a single 3.2s in-app face video guided by liveness prompts.

3 Process

- The system detects the face, embeds features, and quality-checks sampled frames.

4 Done

- 5 independent 128-d templates are generated and saved per worker.



The **5-Template Advantage**: Storing 5 templates per person lifts the top-1 accuracy from 95.49% to an impressive 97.21% on the LFW attendance-like test.

Offline Integrity and State Management

The system functions fully offline for unlimited blackout periods without degrading performance.

Data Minimization

Only numeric 128-d embeddings and metadata persist locally. **Absolutely no face photos** are stored on the device at any time.

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Immutability

Local records carry **client-generated IDs**. This ensures the trail remains **auditable** at the backend and prevents database corruption.

Idempotent Architecture

When network connectivity fluctuates, **sync retries** cannot **accidentally** duplicate attendance records.



The Lifecycle: Reconcile and Purge

DataLake3Auth respects the brief's strict data lifecycle requirements via a background or manual sync queue.



1. **Queue Locally:** Verified attendance records sit in an offline queue.
2. **Sync to AWS:** Upon reconnection, an API call is made uploading pending items.
3. **Validate:** The system reports Synced, Failed, or Purged status back to the dashboard.
4. **Purge Local Copy:** Confirmed, successfully ingested records are immediately permanently deleted from the device's local storage to maintain a clean security footprint.

Backend Validation: Proven Sync Path

The end-to-end sync path is fully validated against standard AWS infrastructure.

- **API Gateway:** Receives the device's sync POST payload securely.
- **AWS Lambda:** The datalake3auth-sync-ingest function successfully logs and writes.
- **Amazon DynamoDB:** DataLake3AuthSyncRecords persists the items.

CloudWatch Sync Proof

Timestamp: 2026-06-05T05:48:57.990Z

Message: INFO Sync table: attendance_logs

Result: verification_success

DynamoDB Persistence Proof

Table: DataLake3AuthSyncRecords

Stored Partition Key: TABLE#attendance_logs

Developer Experience: Drop-In Integration

Designed as a modular engine ready to embed directly into Datalake 3.0 with minimal friction.

The JS-Facing API

```
enroll(worker, video);  
  
verify(video) > {workerId, confidence, margin, liveness};  
  
sync();
```

Deployment:

Add the module's screens to the host navigator. Link the native ONNX bridge for iOS/Android, and bundle the 16.14 MB assets.

Flexibility:

Keep the provided validated AWS stack or simply point the sync() endpoint to Datalake's existing backend.

Enterprise Privacy by Construction



Anti-Spoofing First

In-app-only capture combined with active prompts and temporal fusion neutralizes presentation attacks at the hardware edge.



Absolute Minimization

Zero face photos persist. The system translates identity into mathematical vectors locally and destroys the source material instantly.

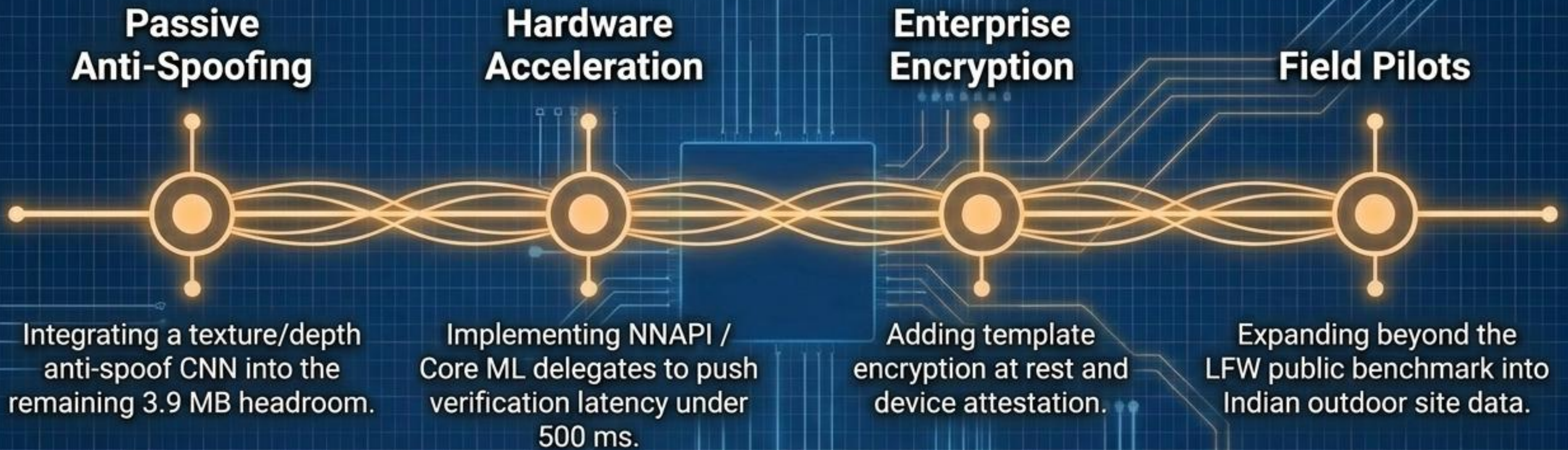


Cryptographic Auditability

Immutability of local records combined with idempotent sync guarantees that the attendance audit trail perfectly consistent, even across highly unstable network links.

The Horizon: Utilizing the Headroom

The 3.86 MB of space saved in the footprint budget allows for immediate, high-value roadmap expansions without breaking Hackathon 7.0 parameters:



Deliverables Ready for Review:

Full Source Code | Technical Report (PDF) | AWS Validation Evidence
github.com/divyaman-pal/Assignment

